EXP No : 12 **Study and implement the Hierarchical Clustering for data grouping**
Date :

Aim:

To study and implement Hierarchical Clustering for grouping data using Agglomerative Clustering techniques.

Procedure: Agglomerative Clustering – Single Linkage

1. Select Agglomerative Clustering with Single Linkage.
2. Enter the required data values in the given table.
3. Calculate the Euclidean distance for each pair of data points using:

$$d = \sqrt{(X_1 - X_2)^2 + (Y_1 - Y_2)^2}$$

4. Enter the calculated distance in the corresponding cell of the distance matrix.
5. Repeat the process until the complete distance matrix is obtained.
6. Click the MIN button to identify the clusters having the smallest distance.
7. Click MERGE to combine the selected clusters.
8. Recalculate the distances using the Single Linkage criterion.
9. The distance between two clusters is the minimum distance between any two points belonging to the clusters.
10. $D(C_1, C_2) = \min d\left(x_i, x_j\right)$
11. Repeat the MIN → MERGE process until only one cluster remains.
12. Click NEXT to display the dendrogram.
13. Click PLOT to plot the final dendrogram.

OBSERVATION:(Simulation)

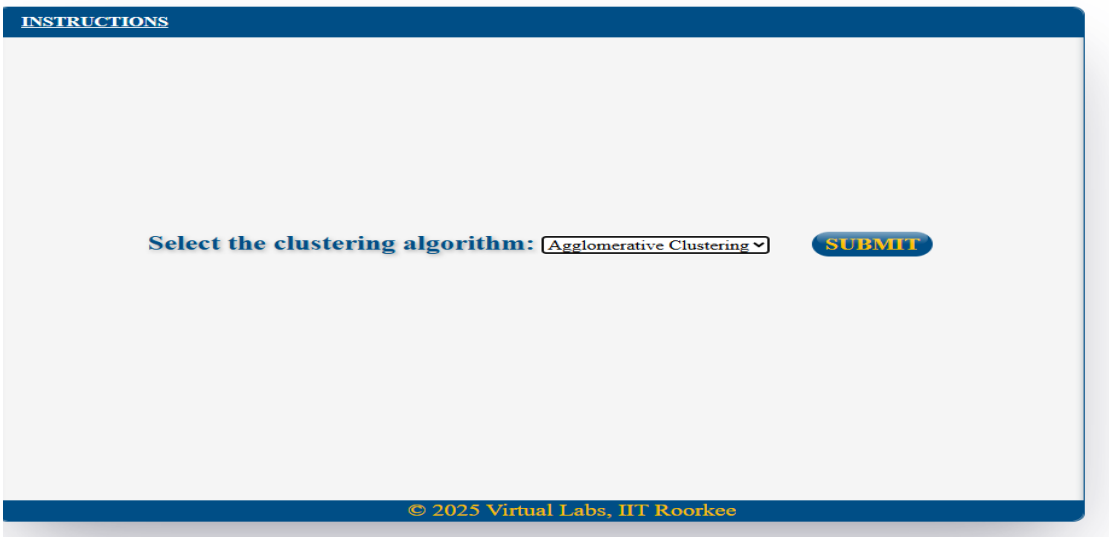


Fig 1: *Selecting the clustering algorithm (Agglomerative Clustering)*

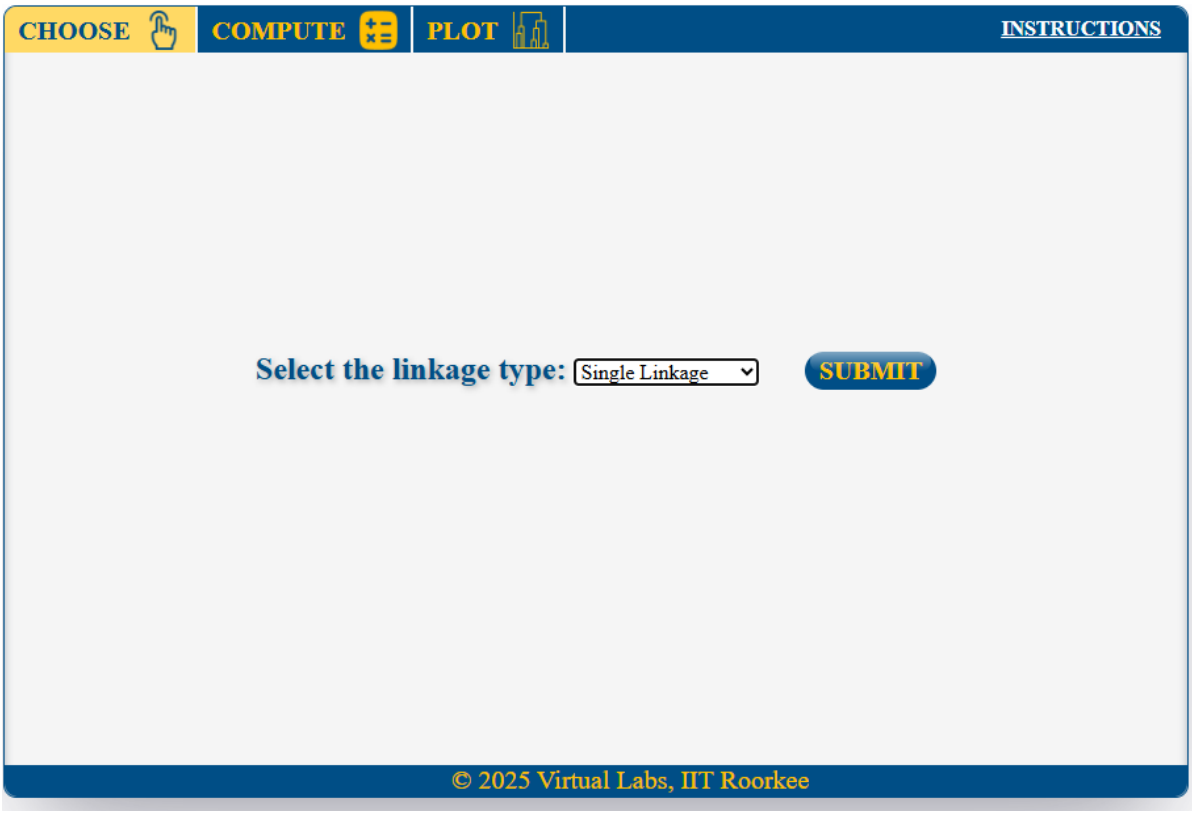


Fig 2: *Selecting the linkage type (Single Linkage)*

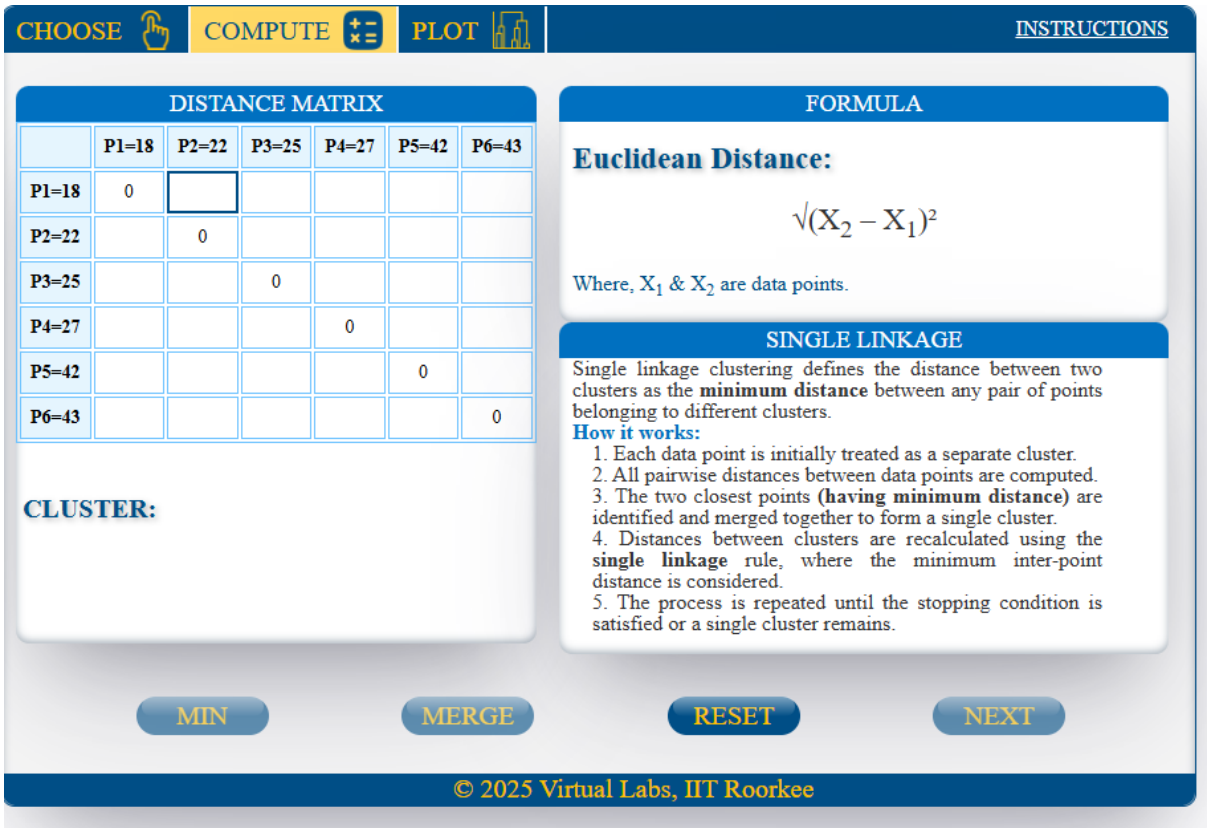


Fig 3: *Initial distance matrix and formula display*

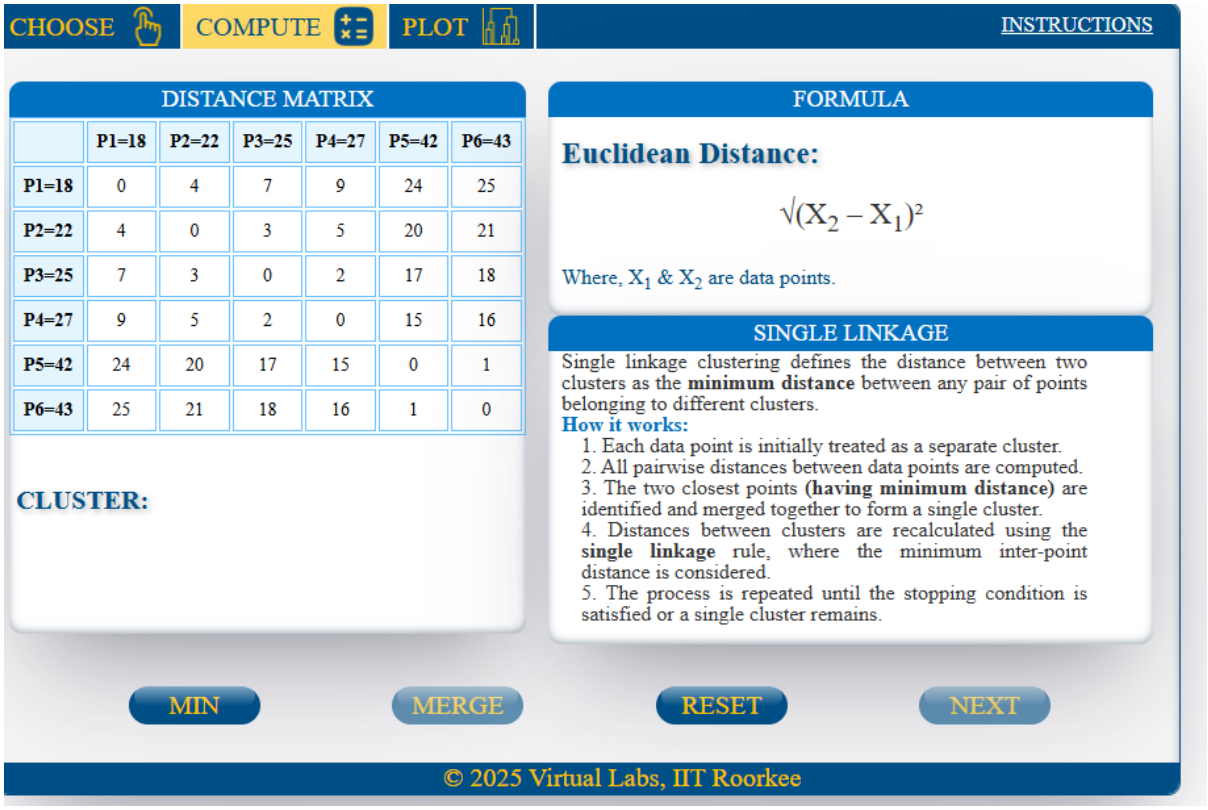


Fig 4: *Calculated distance matrix for all data points*

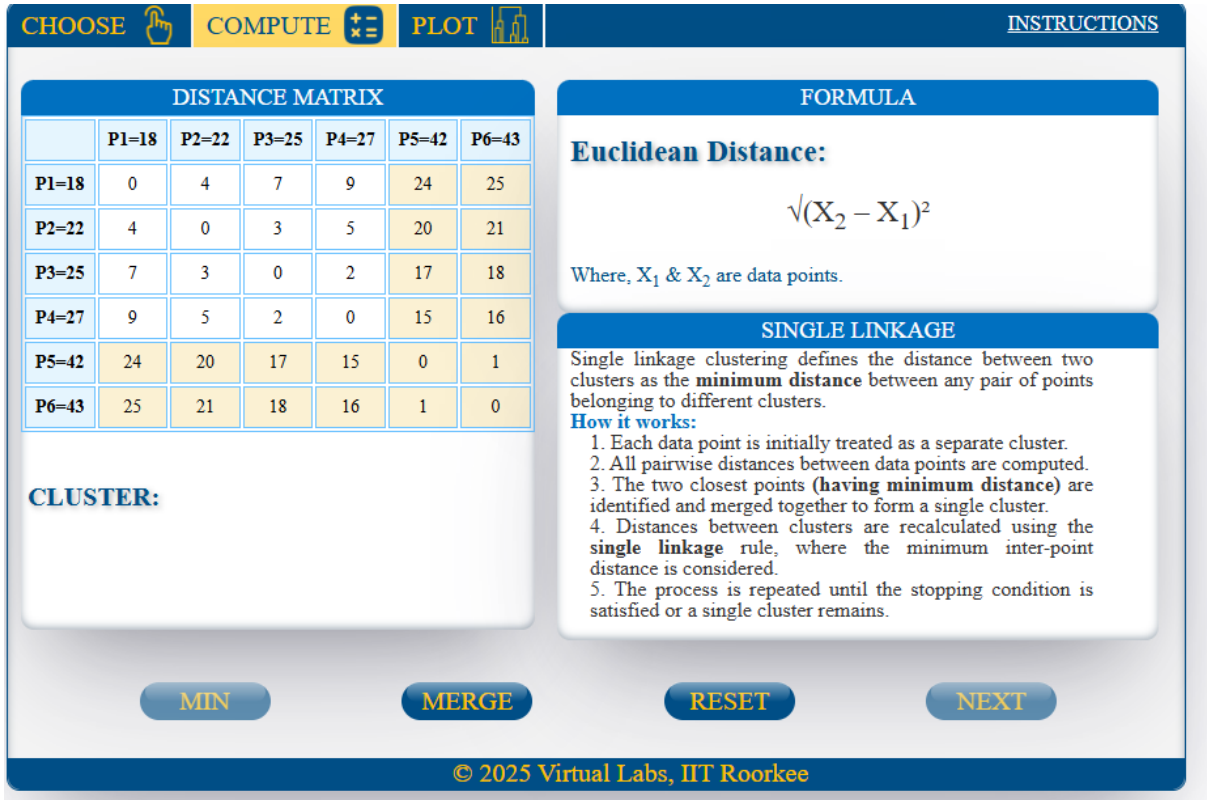


Fig 5: *Finding the minimum distance in the distance matrix*

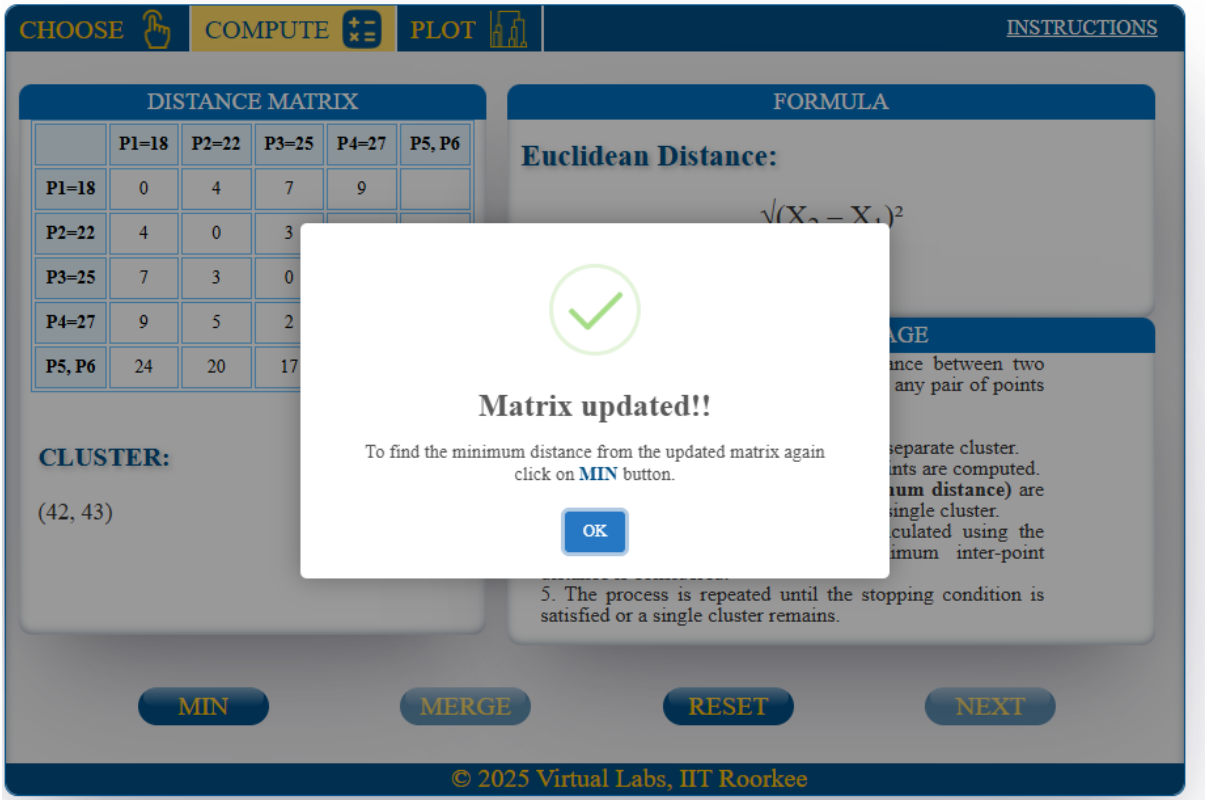


Fig 6: *Popup confirmation after updating the distance matrix for cluster(42, 43)*

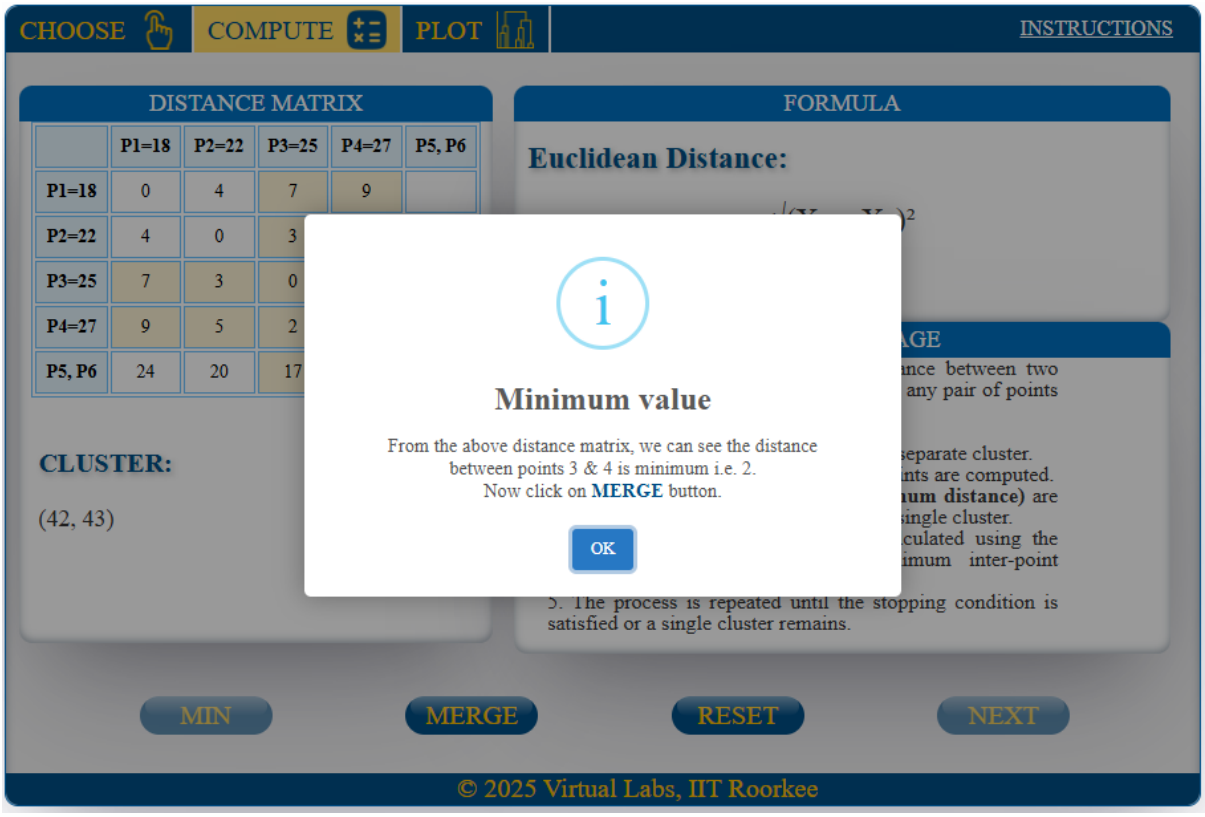


Fig 7: *Highlighting minimum distance between points 3 and 4*

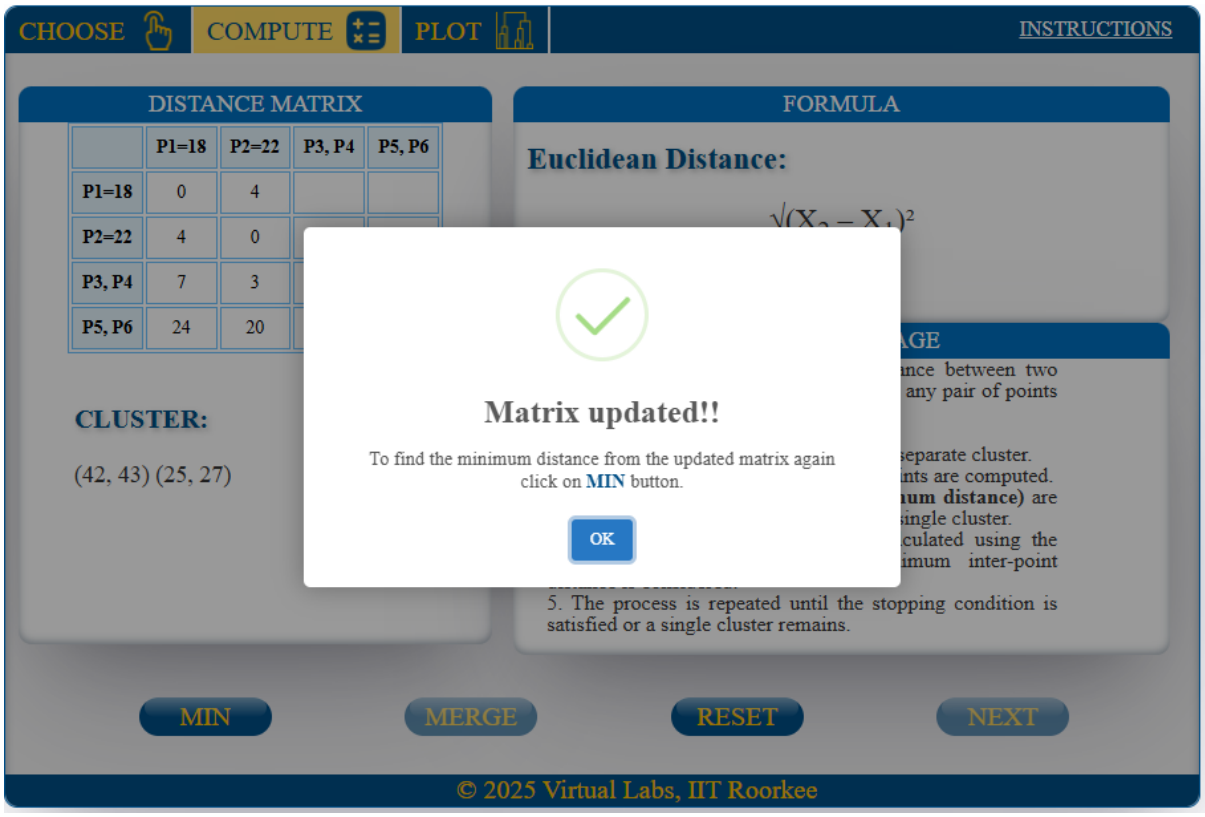


Fig 8: *Popup confirmation after updating matrix for cluster (25, 27)*

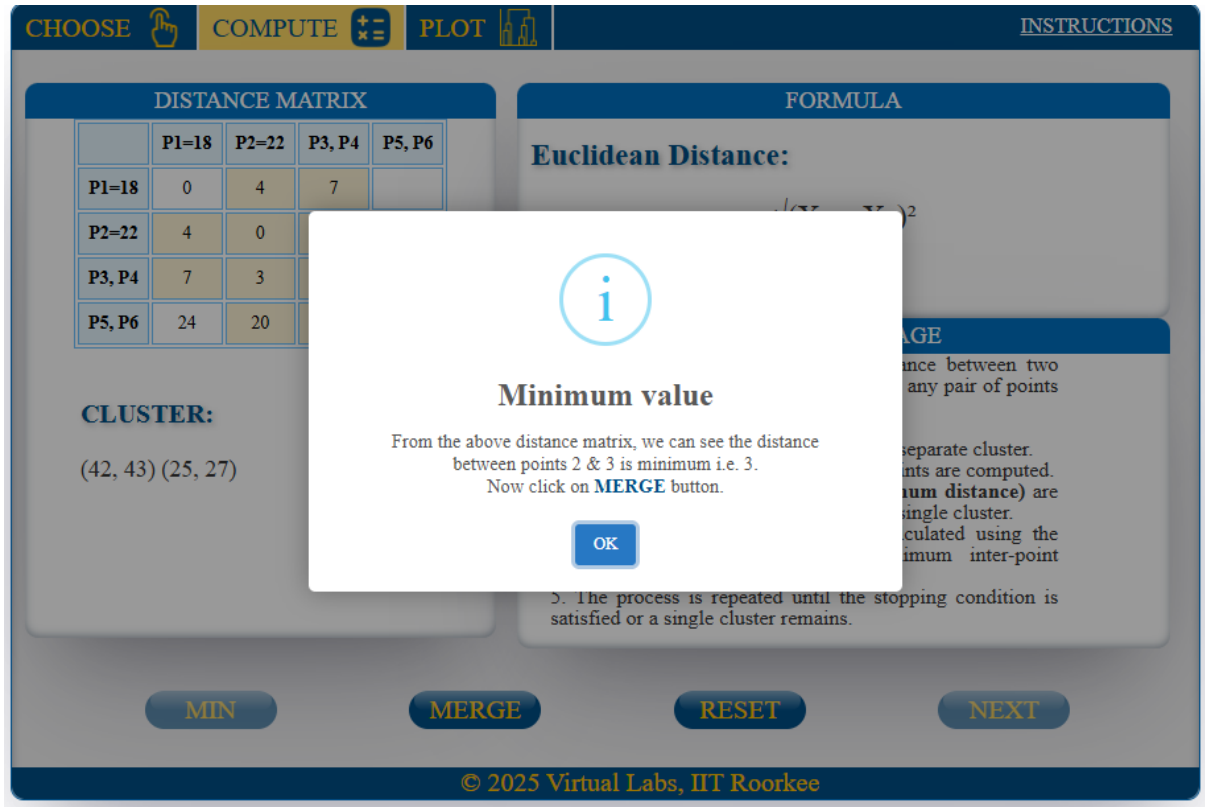


Fig 9: *Highlighting minimum distance between point 2 and cluster (25, 27)*

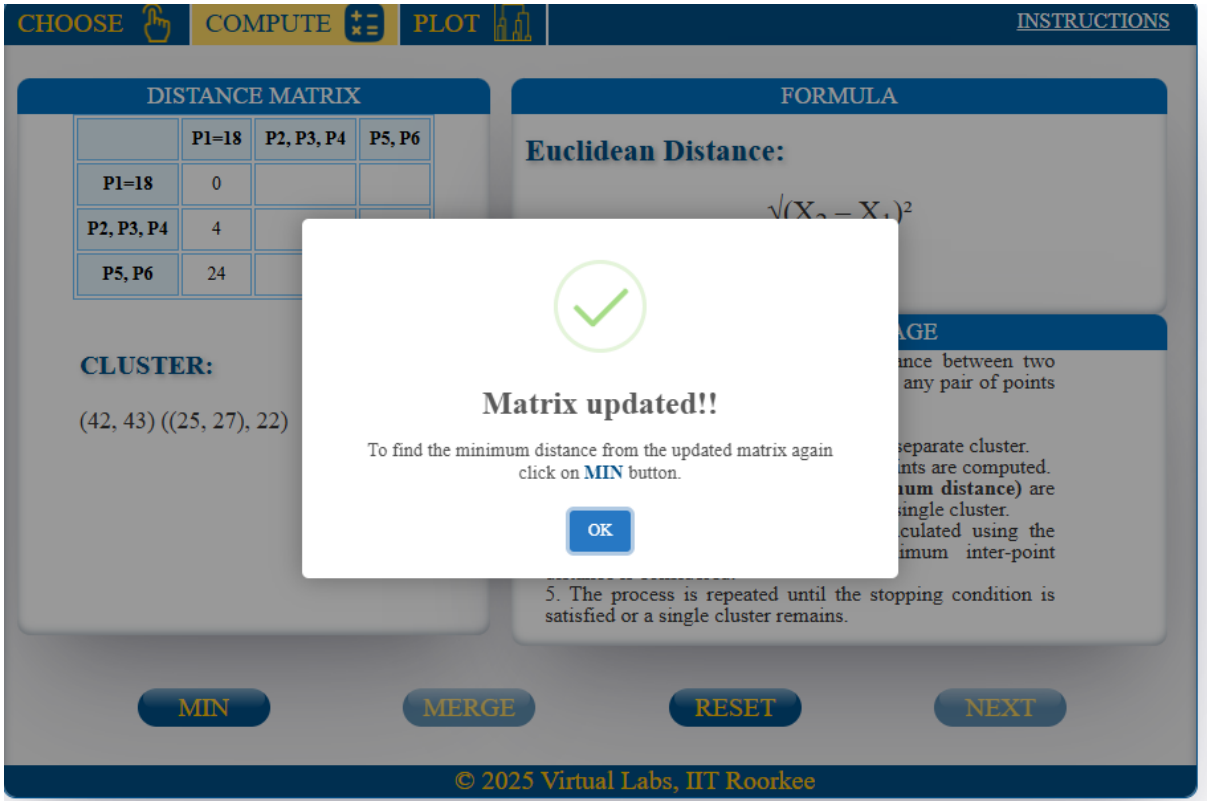


Fig 10: *Popup confirmation after merging cluster ((25, 27), 22)*

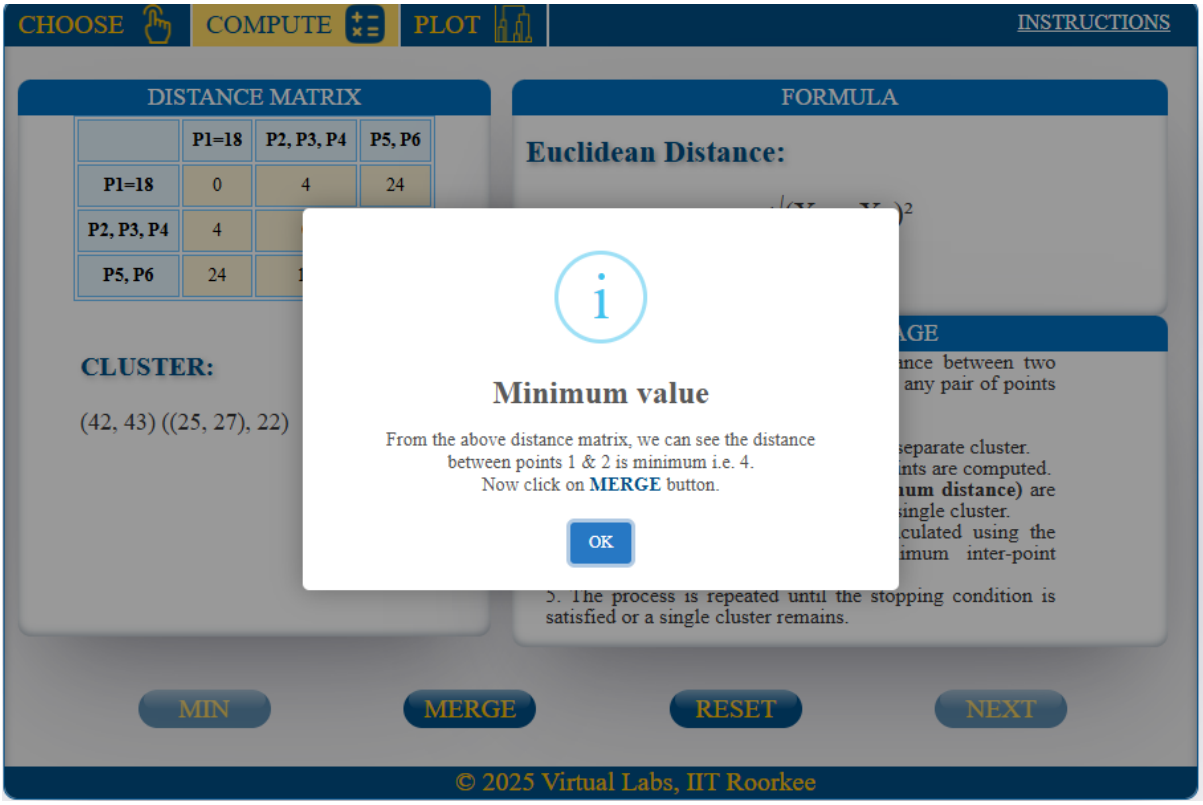


Fig 11: *Highlighting minimum distance between point 1 and cluster((25, 27),22)*

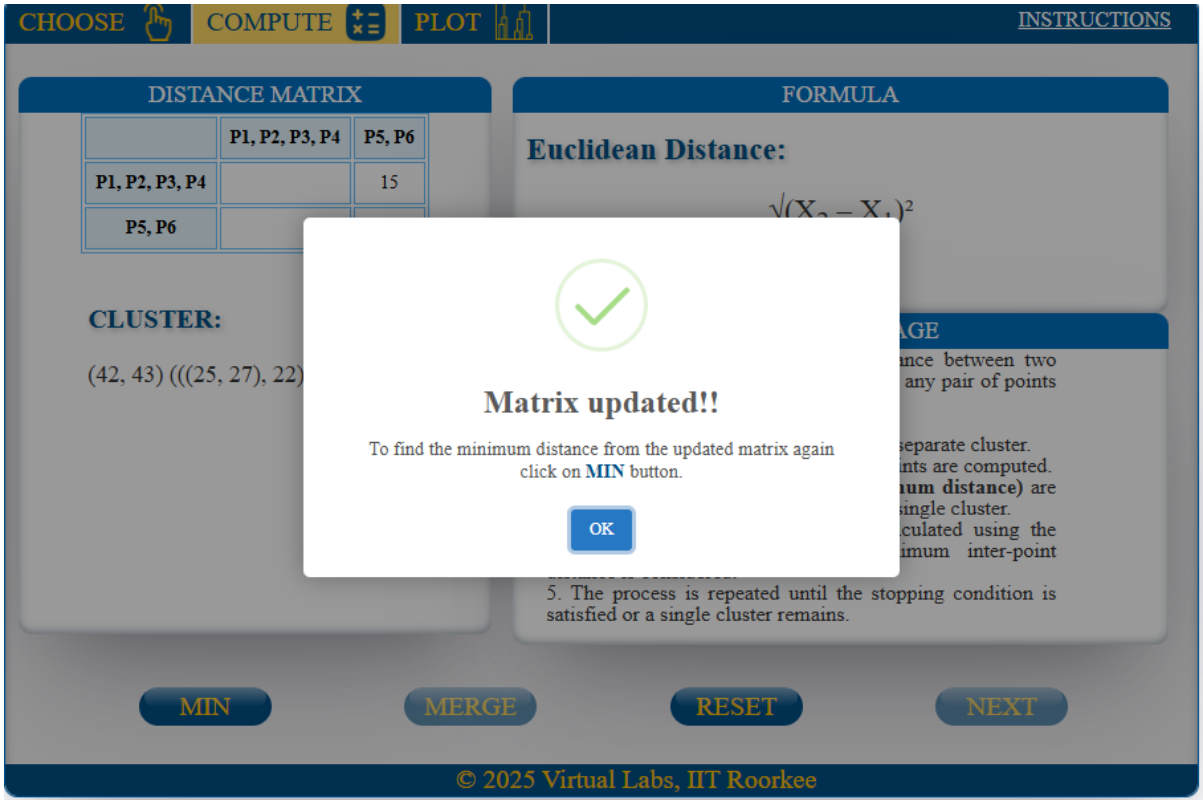


Fig 12: *Popup confirmation after updating matrix for cluster(((25, 27), 22), 18)*

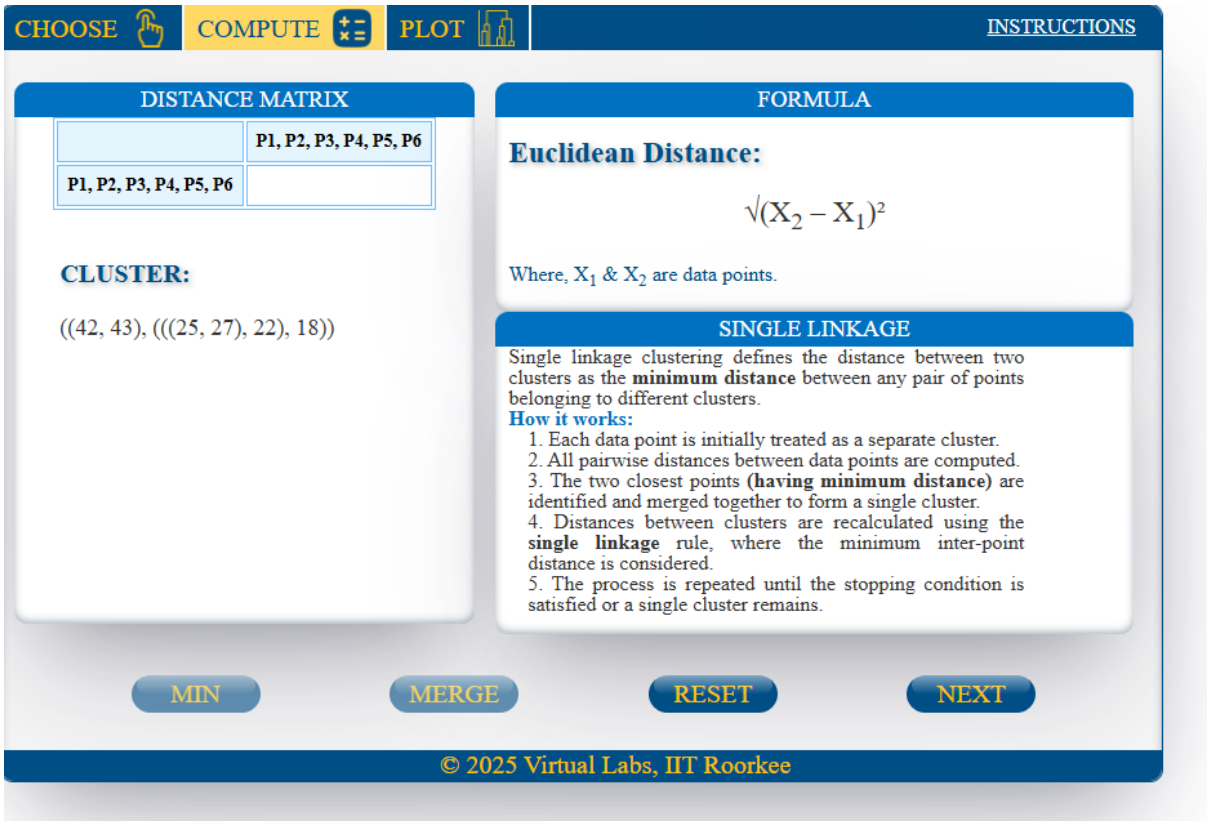


Fig 13: *Final distance matrix calculation before total cluster merge*



Fig 14: *Prompt message asking to plot the dendrogram*



Fig 15: *Viewing selected data points list*

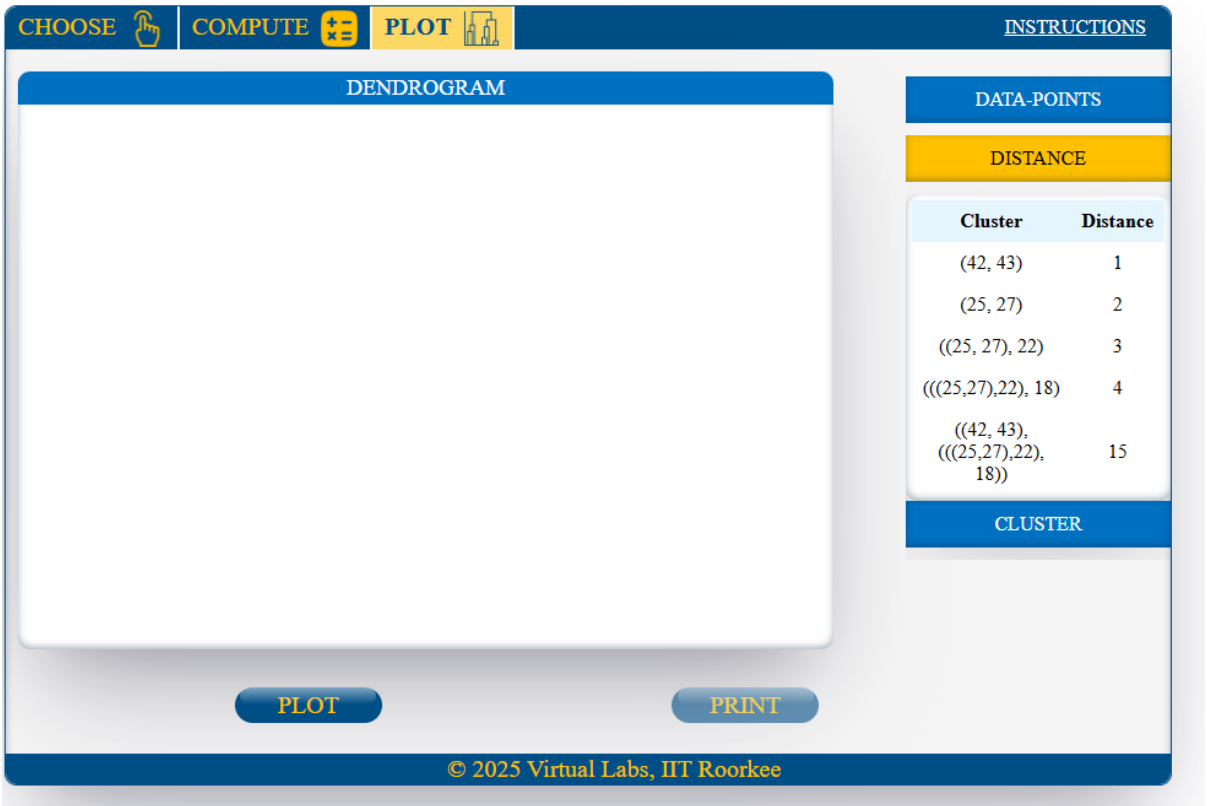


Fig 16: *Viewing cluster distance values table*

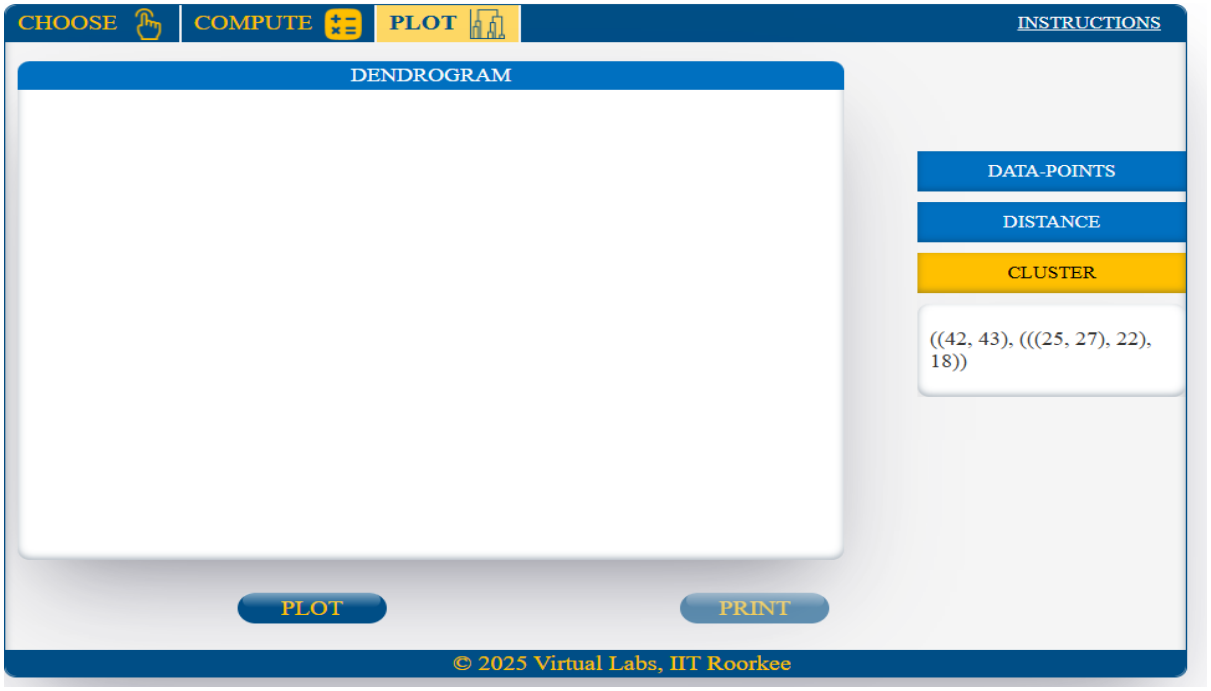


Fig 17: *Viewing final merged cluster output string*

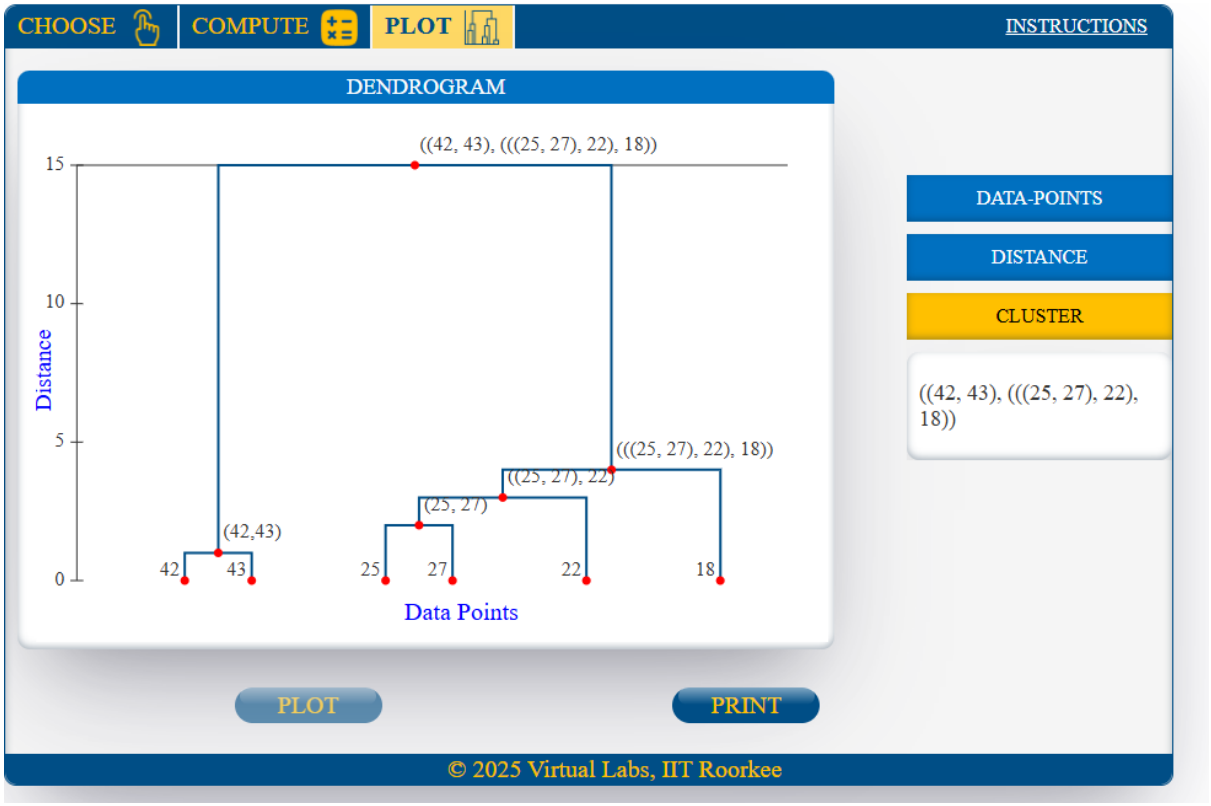


Fig 18: *Final plotted dendrogram showing full hierarchical cluster tree*

Simulation link*:* https://ml-iitr.vlabs.ac.in/exp/hierarchical-clustering/simulation.html

Result:

Thus, Hierarchical Clustering was studied and implemented using Agglomerative Clustering - Single Linkage. The clusters were successfully formed and represented using dendrograms based clustering.